

PolyToken

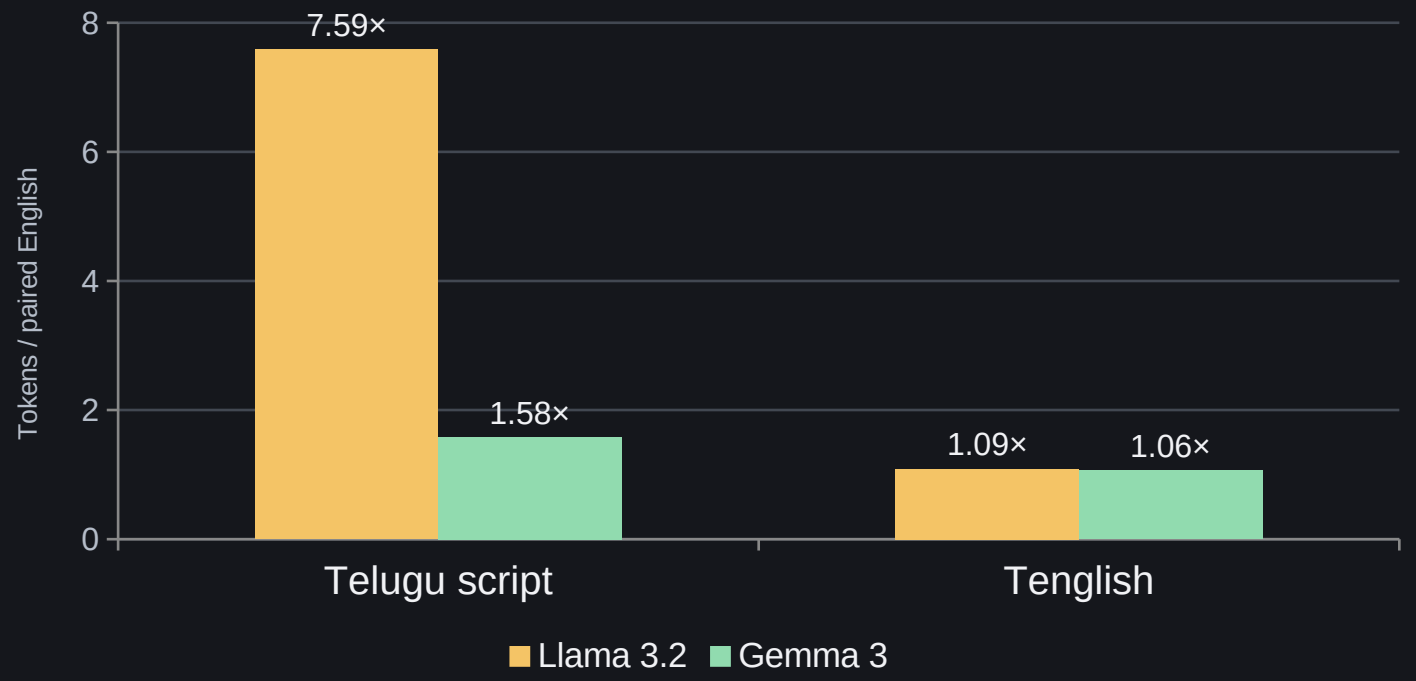
The model-fit advisor for Indian-language AI

Which model fits your users — and why?

Mohan Chelluru

THE PROBLEM

A price sheet cannot see how your users write.



7.59× → 1.09×

Same language.
Different script.
Different token bill.

Ratios to English, not a quality score.

Source: logs/sample/experiment01/fertility.csv + fertility.md · 18 paired reference sentences; not company data.

WHO IT IS FOR

One decision. Four very different workloads.

APP SUPPORT

Samudra Telecom

Tenglish + romanized Telugu

Longer chat history

VOICE CARE

Swara Care

Hindi in Devanagari

Long spoken answers

VOICE ADVISORY

Raitu Mitra

Telugu script from ASR

Farmers and rural vocabulary

FINTECH CHAT

Paisa Saathi

Short Hinglish messages

English product terms

“Which model fits our users and our budget — and why?”

Sources: logs/sample/agent_{samudra,swara,raitu,paisa}_model_fit_report.json · All four companies are fictional.

The agent explains the path to its answer.

01

Understand

Script mix, channel, history and answers

→

02

Measure

Paired token counts through Zetaris SQL

→

03

Compare

Measured counts × dated provider quotes

→

04

Check

Recompute claims; drop unsupported ones

→

05

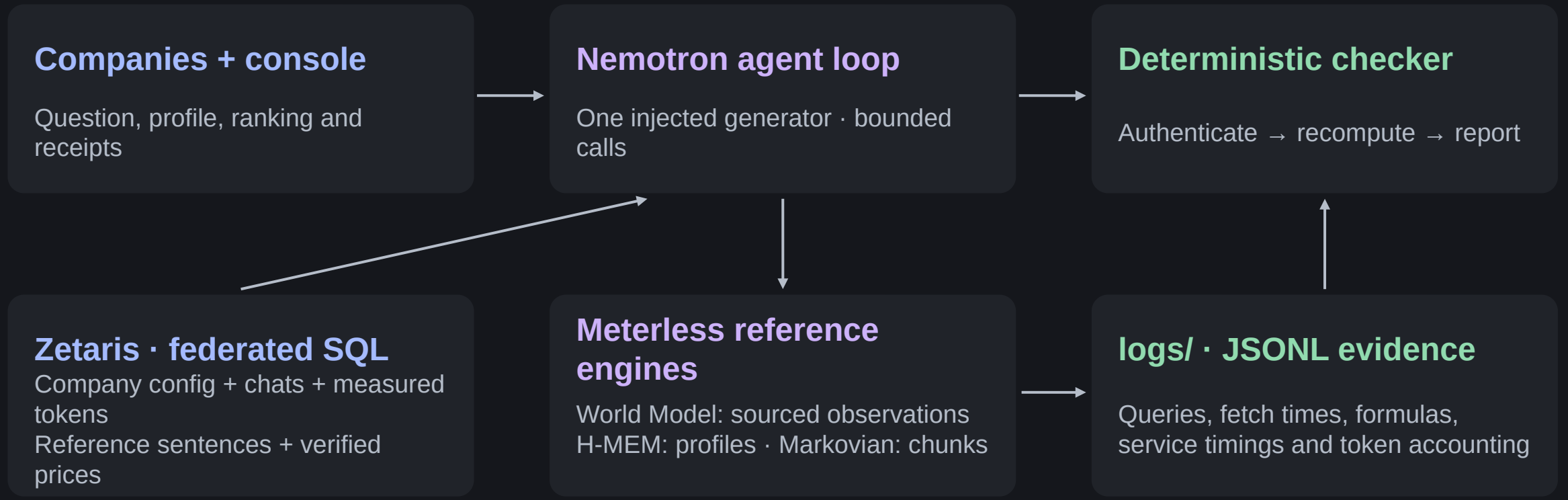
Explain

Ranked cost fit, receipts and limits

Nemotron plans each bounded step. Meterless state preserves its evidence.

Sources: src/polytoken/agent/ · src/polytoken/model_fit.py · docs/METERLESS.md · logs/sample/model_fit_v1_audit_*.jsonl

One reasoning loop. Evidence at every boundary.



Sources: src/polytoken/{llm,zetaris,checker,model_fit}.py · src/polytoken/meterless/ · sql/token_ratio_by_part.sql

The lowest unit price costs 2.61× the best fit.

BEST CONTENT-COST FIT

Gemma 3 4B

₹139

1.35× English tokens

CHEAPEST PER TOKEN

Llama 3.1 8B

₹363

5.03× English tokens

Per month at 80,000 conversations · content estimates, not invoices

Voice latency unverified. Quality: verify on your sample.

Source: logs/sample/agent_raitu_model_fit_report.json · cost + token_ratio + monthly_conversations claims; data/reference/fx.json

The best fit follows the workload.

Company / volume per month	Best content-cost fit	Lowest unit-price model	× best fit
Samudra Telecom 2,00,000 conversations	Llama 3.1 8B ₹176 / month	Llama 3.1 8B ₹176 / month	1.00×
Swara Care 1,20,000 conversations	Gemma 3 4B ₹275 / month	Llama 3.1 8B ₹359 / month	1.31×
Raitu Mitra 80,000 conversations	Gemma 3 4B ₹139 / month	Llama 3.1 8B ₹363 / month	2.61×
Paisa Saathi 3,00,000 conversations	Llama 3.1 8B ₹154 / month	Llama 3.1 8B ₹154 / month	1.00×

Chat mixtures favor Llama. Native-script voice workloads favor Gemma on cost.

Voice deployment decisions await matching latency evidence. All quality remains to verify.

Sources: logs/sample/agent_{samudra,swara,raitu,paisa}_model_fit_report.json; data/reference/fx.json (23 Oct 2026).

A claim earns its place in the report.

29

Meterless conformance checks

241

Tests passed · 8 live tests skipped

12 / 12

Pre-registered addendum checks

Receipt = formula + inputs + query + source + fetch time

Unsupported claims are dropped. Missing timing is not a passed voice constraint.

Sources: docs/METERLESS.md · logs/sample/fit_polish_validation.jsonl · logs/sample/model_fit_score.jsonl · video_deck_validation.jsonl

SPONSOR TECHNOLOGY

Data. Memory. Reasoning. Each with a clear job.

THE DATA LAYER

Zetaris

Federated SQL over conversations, configuration, measurements, reference text and prices.

THE STATE ARCHITECTURES

Meterless

Our independent implementations: World Model, H-MEM and Markovian.

THE AGENT'S BRAIN

NVIDIA

Nemotron planning and judging through one configured generator, with shared rate limits and budgets.

Cursor was not used. · Codex and Claude assisted development.

Sources: docs/SUBMISSION.md · docs/METERLESS.md · docs/DECLARATIONS.md · source and audit paths in repository.

NEXT EVIDENCE TO EARN

From a cost candidate to a deployment decision.

1	**Native review**	Review translations and replace fictional templates with consented samples.
2	**Serving evidence**	Measure the exact model, provider and voice workload against its target.
3	**Controlled quality**	Run paired controls and assess meaning before switching models.

PolyToken. Every token accounted for.

Sources: checked model-fit reports, “limitations” and “serving_note”; docs/CONTRACT.md · Proposed next work, not completed results.